Q&A

Things You Wanted to Know!

Q1. What is the difference between an Arduino development board and a single microcontroller chip? Do I need any software programming for Arduino? Please suggest a suitable book for Arduino.

Raman Chidambaram

A1. The difference between an Arduino development board and a microcontroller chip is that the development board contains almost every essential part needed to start a project on the board itself. It has a microcontroller chip and ports for connection or extension (refer Fig. 1). You can program the chip with inbuilt examples available on Arduino IDE (software) and observe the output right on the board, or by interfacing with external circuits.

On the other hand, for a single microcontroller chip (such as Atmega328 microcontroller as shown in Fig. 2), you need a lot of things. First, you need to buy a separate programmer board to program the chip. For building a project or experimentation with the chip, you need a breadboard or veroboard for circuit wiring/connections with other components, a DC power supply for the project, jumper wires, etc. You also need to search for program examples from various sources, books, and the Internet.

Fig. 1: Arduino Uno development board

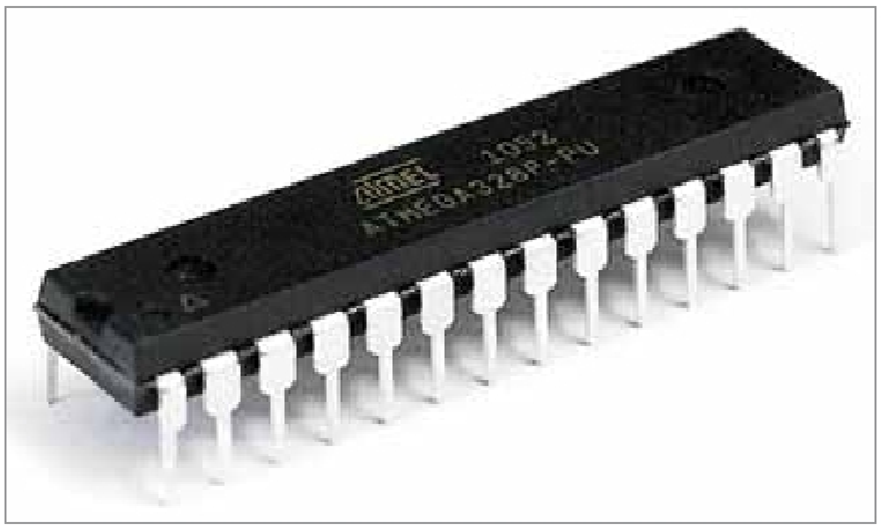

Fig. 2: ATmega328P microcontroller chip

To work with Arduino, you need to work with the Arduino software programming language.

You can buy Arduino book (E-book) from various online stores such as *www.magzter.com*, *www.amazon.in* and others.

Q2. What are the different types of BLDC motors available in India? Where can I get a list of DC motor manufacturers in India?

L. Palaniyappan

A2. A brushless DC motor (BLDC) is basically a synchronous DC motor powered by direct current (DC) and consists of two main parts: a stator and a rotor. Broadly, there are two types of BLDC motors based on the physical design—inner rotor design and outer rotor design. The inner rotor design is a conventional design in which the rotor is located at the core (centre), and the stator coil winding surrounds it. In the outer rotor design configuration, the stator windings are located at the core, while the rotor carrying permanent magnets surrounds the stator.

Again, there are two types of BLDC based on the sensors—sensor-based BLDC and sensorless BLCD. A sensor-based BLDC motor is more accurate but costly, and is used for a specific application.

BLDC motors are available in three configurations: single-phase, two-phase, and three-phase. Out of these, the three-phase BLDC is the most common. All these motors are easily available in India and online shops.

You can get the list of Indian manufacturers, including DC motors, BLDC, synchronous motors, etc, from the *eleb2b.com* website.

Answers compiled by Sani Theo, technical editor. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: **editsec@efy.in**) and should include name and address of the sender.